Dosimetry

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1 Radiation Dose (Dosage), Dosimetry, and Dosimetry Methods

1.1 Radiation Dose

Radiation dose is a quantitative measure of the energy deposited by ionizing radiation in matter, particularly biological tissues.

1.1.1 Exposure

$$X = \frac{dQ}{dm} \tag{1}$$

where Q is electric charge and m is the mass of air. The SI unit is C/kg, with

$$1 \text{ R} = 2.58 \times 10^{-4}\,\text{C/kg}.$$

1.1.2 Absorbed Dose

$$D = \frac{dE}{dm} \tag{2}$$

The SI unit is the Gray:

$$1 \text{ Gy} = 1 \text{ J/kg} = 100 \text{ rad}.$$

1.1.3 Equivalent Dose

$$H_T = Dw_R \tag{3}$$

1.1.4 Effective Dose

$$E = \sum_T w_T H_T \tag{4}$$

Radiation	w_R
X-rays	1
Gamma rays	1
Beta particles	1
Protons	2
Alpha particles	20
Neutrons	5–20

Table 1: Radiation weighting factors.

2 Radiation Dosimetry

Radiation dosimetry is the science of measuring absorbed radiation dose.

Applications include radiation therapy, diagnostic radiology, nuclear medicine, industrial radiography, environmental monitoring, occupational protection, and space radiation monitoring.

3 Types of Dosimeters

3.1 Film Badge

Film badges darken in proportion to radiation exposure due to silver bromide ionization.

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Radiation
   |
Film
   |
Latent Image
   |
Development
   |
Dose
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Advantages: inexpensive, permanent record. Disadvantages: single-use, affected by heat and humidity.

3.2 Thermoluminescent Dosimeter (TLD)

Heating releases trapped electrons from LiF, $CaSO_4$, or CaF_2 crystals.

$$L \propto D$$

3.3 Optically Stimulated Luminescence (OSL)

Laser light releases trapped electrons from Al_2O_3:C crystals. The emitted light is proportional to dose.

3.4 Ionization Chamber

$$I = \frac{Q}{t} \tag{5}$$

Gas ionization current is measured with an electrometer.

3.5 Geiger–Müller Counter

Radiation initiates a Townsend avalanche producing detectable electrical pulses.

3.6 Semiconductor Dosimeters

Radiation creates electron-hole pairs in silicon or diamond.

$$Q \propto D$$

3.7 Chemical Dosimeters

Fricke dosimeters measure:

$$\mathrm{Fe}^{2+} \rightarrow \mathrm{Fe}^{3+}$$

3.8 Alanine Dosimeters

Alanine crystals form stable free radicals measured by ESR spectroscopy.

4 Comparison of Dosimetry Methods

Dosimeter	Principle	Readout	Use
Film Badge	Film darkening	Delayed	Personnel
TLD	Thermoluminescence	Delayed	Medical
OSL	Optical luminescence	Delayed	Personnel
Ionization Chamber	Gas ionization	Real time	Calibration
GM Counter	Gas avalanche	Real time	Surveys
Semiconductor	Electron-hole pairs	Real time	Radiotherapy
Fricke	Chemical oxidation	Delayed	High dose
Alanine	ESR	Delayed	Industrial

Table 2: Comparison of dosimetry methods.

5 Applications of Dosimetry

- Diagnostic radiology
- Radiation therapy
- Nuclear medicine
- Occupational monitoring
- Environmental monitoring
- Industrial irradiation
- Space radiation monitoring